Shabir Hussain

ce23btech11047@iith.ac.in | kachoshabir shah@gmail.com | +91 8899671463 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.71
boys higher secondary school baroo kargil	Class XII (jkbose)	2022	80.60%
high school lankerchy	Class X (jkbose)	2020	7.80%

WORK EXPERIENCE

Machine Learning Intern | Avartan Labs pvt ltd

- Developed a **next-hour waste intake forecasting system** for **14 material categories** using hourly operational data.
- Engineered **20+ temporal and operational features,** including lag features, rolling statistics, and cyclical time encodings to improve prediction accuracy.
- Built and compared **Random Forest, Gradient Boosting Ensemble, ARIMAX, and Bidirectional LSTM** models using walk-forward validation for time-series forecasting.
- Evaluated model performance with **MAE, RMSE, R^2, MAPE, sMAPE, and Pearson Correlation** to identify the most accurate forecasting model.

PROJECTS

Expense Management System

- Built a modular **C++ expense-management system** using **OOP, encapsulation, and class-based architecture**.
- Implemented **Equal, Exact, and Percentage splits** with expense and contribution validation.
- Designed a **greedy debt-settlement algorithm** to compute net balances and minimize settlement transactions.
- Used **STL vectors, hash maps, and modular components** for efficient user, group, expense, and balance management.

Distributed File System

- Designed and implemented a **peer-to-peer distributed file storage system** in Go using **TCP socket programming**, enabling decentralized file storage, retrieval, and concurrent communication across multiple nodes.
- Built **Content Addressable Storage (SHA-1)** with **AES encryption** and **Gob-based message serialization** to provide secure, deduplicated storage and efficient data integrity verification.
- Developed **automatic file replication, bootstrap-based peer discovery,** and **synchronized deletion** mechanisms to ensure **fault tolerance, high availability,** and **consistency** across distributed storage nodes.

Geospatial Machine Learning for Irrigation *(Prof* Phanindra K.B.V.N.*)*

- Developed a **cloud-native geospatial ML framework** using **JavaScript (Google Earth Engine) and Python**, integrating HLS, Landsat 8/9, Cartosat-2E, and SRTM DEM.
- Implemented **TsHARP spatial disaggregation, Random Forest LST prediction, HS-FRAG field segmentation, and SEBAL energy-balance modeling**.
- Generated **30 m synthetic LST at 3-day intervals** with **R 0.904**, enabling field-scale **ET estimation and irrigation** scheduling.

AI Chatbot with RAG & Tool Calling

- Developed an **AI chatbot** using **LangGraph** and **Llama 3**, supporting real-time streaming conversations.
- Implemented **RAG** with **FAISS** to answer questions from uploaded PDF documents.
- Integrated **tool calling** for calculator, web search, and stock price retrieval within the chatbot.
- Built a **modular Streamlit application with SQLite-based** persistent multi-session chat memory.

SKILLS

Programming Languages: C, C++, Python, JavaScript, SQL.

CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN), System Design.

Tools & Platforms: Git/GitHub, VS Code, MySQL,Postman.

Machine Learning: NumPy, Pandas, Scikit-learn, TensorFlow, Machine Learning, Deep Learning

RELEVANT COURSES

Mathematics: Calculus 1, Calculus 2, Elementary Linear Algebra, Differential Equations, Transform Techniques, Probability, Complex Variables, Statistics.

Programming: Introduction to Programming, Data Structures and Algorithms.

Extra Courses: Spoken Japanese, Communication Skills.

SCHOLASTIC ACHIEVEMENTS

- Solved **1000+** DSA problems across coding platforms, including GeeksforGeeks, LeetCode and Codeforces.

POSITIONS OF RESPONSIBILITY

Transport Coordinator - Inter IIT Sports Meet 2025.

Placement Coordinator - Office of Career Services IIT Hyderabad.

EXTRACURRICULAR

- Participated in the Under-18 State-Level Cricket Championship (2021-22).
- Gold Medalist (2025) and Bronze Medalist (2024) in Inter-Departmental Cricket Championship.
- NSO (National Sports Organization) Cricket Member.
- Under-16 Cricket & Volleyball at District Level (2019-20)